① Charge is quantized. q.

$1e = 1.6 \times 10^{-19}\ C$.

↳ Smallest amount of charge, cannot be divided.

② Coulomb's law.

VECTOR SUPER POSITION

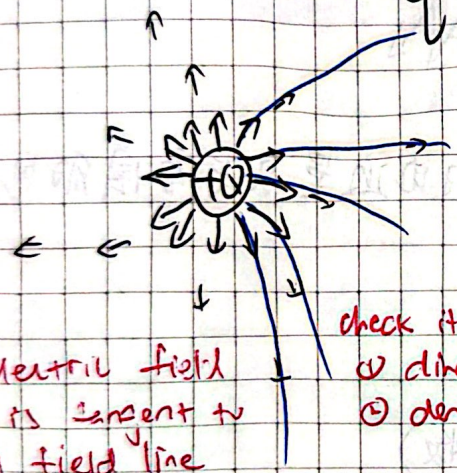
→ like charges repel
unlike charges attract.

$$\vec{F} = k\left|\frac{q_1 q_2}{r^2}\right| \qquad k = 9 \times 10^9\ Nm^2/C^2$$

$$k = \frac{1}{4\pi\varepsilon_0},\ \varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$$

③ Electric Field.

$$\vec{E} = \frac{\vec{F}}{q}$$, the force a unit charge will experience in electric field

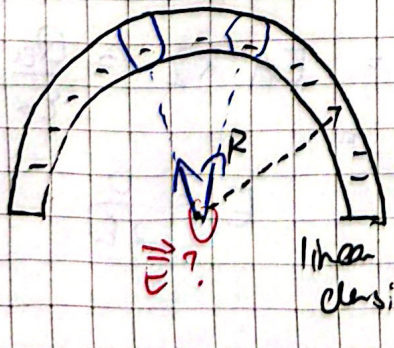


$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

check its
① direction
② density (strength)

* electric field vector is tangent to any field line drawn.

Ex.



due to symmetry, we should only calculate the sum of field in vertical direction due to cancelled out horizontal field.

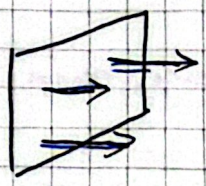
$$E = \int_0^{\pi} \frac{k\,dQ}{R^2}\cdot\sin\theta = \int_0^{\pi} \frac{k\sin\theta}{R}\lambda\,d\theta$$

$$dQ = \lambda\,dl = \lambda R\,d\theta = \left[-\frac{k\lambda}{R}\cos\theta\right]_0^{\pi}$$

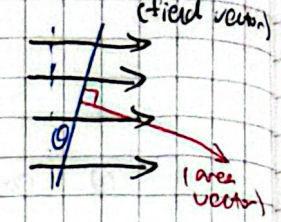
$$\frac{2k\lambda}{R} = \frac{\lambda}{2\pi\varepsilon_0 R}$$

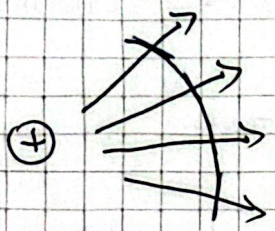
① GAUSS LAW.

1. Electric Flux: amount of electric field passing through surface.



$$\Phi_E = \vec{E} \cdot \vec{A}$$
$$= EA\cos\theta$$

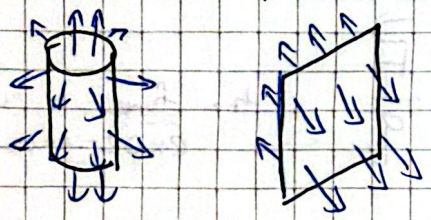




$$\Phi_E = \int d\Phi_E$$
$$= \int E\,dA.$$

2. Symmetry of electric field.

↳ 从何对称去推出电场对称 | 垂直于平面等.

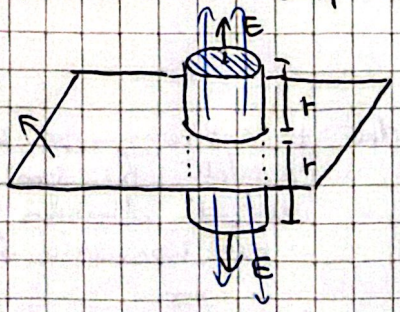


3. GAUSS LAW: 闭合曲面的电通量只与包围的电量有关

$$\Phi_{ex} = \oint E\,dA = \frac{Q_{ex}}{\varepsilon_0}.$$

4. APPLICANCE. (某个平行板)

面密度 σ



by symmetry, electric field of the plate is perpendicular the the plate its self.

$$\Phi_{ex} = EA \times 2 = \frac{Q_{ex}}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

$U_B - U_A$



④ conductor and insulator

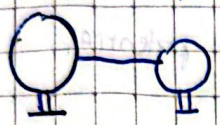
① insulator does not allow charge to move freely.

② conductors allow charge to move freely

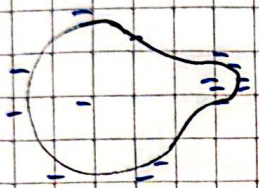


↳ within a conductor, electric field = 0 because electrons will move until no force (net) exerted to them (electrostatic equilibrium)

$\frac{Q}{R} = \frac{q}{r}$



↳ if conductors are connected, charge will redistribute so that inner electric field equals zero



↳ charge will distribute to the surface, by Gauss law $\vec{E}=0=\frac{0 \text{ enclosed } q}{\varepsilon_0}$.

density of charge depends on radius of curvature.

↳ electric field will ⊥ to surface.

⑤ Electric Potential. ENERGY.

1. by definition of potential energy,

$$\Delta U = -W_{CF}$$

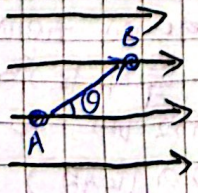
$$\Delta U_E = -W_E$$

$$\therefore W_{A\to B} = \int_A^B \frac{kQ_1Q_2}{r^2}\,dr = -\frac{kQ_1Q_2}{r_B} + \frac{kQ_1Q_2}{r_A}$$

$$U_B - U_A = -W_{A\to B} = \frac{kQ_1Q_2}{r_B} - \frac{kQ_1Q_2}{r_A}$$

$$\therefore U = \frac{kQ_1Q_2}{r}$$ (thus when $r \to \infty$, $U=0$)

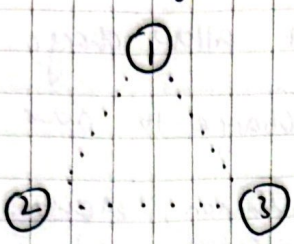
2. in uniform electric field:



$$W = Eqd = Fed. = qEr\cos\theta$$

$$\therefore \Delta U = -Eqd. = -qEr\cos\theta.$$

3. in a system.



$$EPE = \sum U_{\text{All combination}}$$
$$= U_{12} + U_{23} + U_{13}$$

4. Electric Potential

↳ electric potential is electric potential energy per unit charge.

$$V = \frac{U}{q}, \quad \Delta U = \frac{\Delta U}{q}$$

↳ since $F_E = -\frac{dU}{dx}$,

$$\underline{E} = \frac{F}{q} = -\frac{dV}{dx} = -\frac{dU/q}{dx}$$

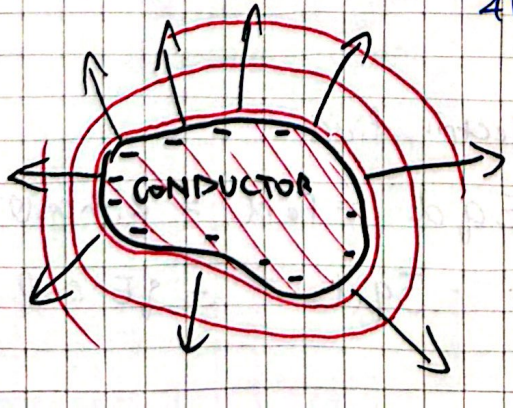
↳
$$V = k\frac{Q}{r}$$

in uniform field, $\Delta V = -Ed$.

↳ scalar quantity.

to calculate the potential in a certain point:

$$V = \frac{1}{4\pi\varepsilon_0}\sum\frac{q_i}{r_i}$$



↳ equipotential surface.

· perpendicular to all electric field lines.

(因为电在等势面移动时不做功，在沿力方向的位移)